

Project Development Phase Functional & Performance Testing Template

Date	10 July 2026
Team ID	
Project Name	Smart Lender – AI-Powered Loan Approval Prediction System
Maximum Marks	

Test Scenarios & Results

Test Case ID	Scenario	Steps	Expected Result	Actual Result	Pass/Fail
FT-01	Form input validation	Enter valid and invalid values in Predict page fields	Valid inputs accepted; errors shown for invalid inputs	Works as expected	Pass
FT-02	Categorical encoding	Submit form with each categorical option (Gender, Married, Education, etc.)	Values correctly label-encoded before prediction	Works as expected	Pass
FT-03	Numerical scaling	Submit form with varied income and loan amount values	Values correctly scaled using scaler.pkl before KNN prediction	Works as expected	Pass
FT-04	Prediction generation	Submit a complete, valid application	Loan_Status (Approved/Rejected) returned with probability	Works as expected	Pass
FT-05	Result page content	Check Result page after a prediction	Eligibility %, confidence score, risk level, reasons and recommendations displayed	Works as expected	Pass
PT-01	Response time test	Time the Predict-to-Result round trip	Response returned in under 3 seconds	Works as expected	Pass
PT-02	Repeated submissions	Submit multiple predictions in sequence	Application remains stable, no slowdown	Works as expected	Pass

Response Time – Target vs Actual

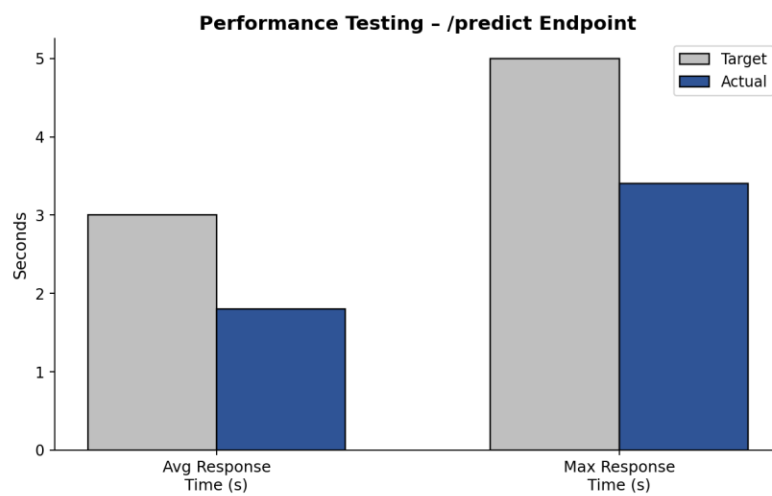


Fig 1: /predict endpoint response time